

MATA37H3 — Week 11

Power Series I: Radius/Interval of Convergence & Function Representations

60 minutes (Pages 1-5, self-paced) • OpenStax *Calculus Volume 2*, §6.1 (Power Series and Functions) — Theorem 6.1 (Convergence of a Power Series), radius/interval of convergence; §6.2 (Properties of Power Series) — Theorem 6.2 (Combining Power Series), Theorem 6.4 (Term-by-Term Differentiation and Integration). MAT137 Unit 14 slides (14.1-14.3: interval of convergence including a “hard” factorial-ratio series; writing functions as power series via geometric-series manipulation) — MAT137 is the University of Toronto’s official exclusion-equivalent of MATA37H3.

DEFINITION Power Series, Radius & Interval of Convergence

A **power series centered at** a is a series of the form

$$\sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \dots$$

By Theorem 6.1, exactly one of the following holds: (i) the series converges only at $x = a$ (**radius of convergence** $R = 0$); (ii) the series converges for **all** real x ($R = \infty$); or (iii) there is a real number $R > 0$ such that the series converges for $|x - a| < R$ and diverges for $|x - a| > R$ — the **interval of convergence** is then $(a - R, a + R)$, possibly with one or both endpoints included, and each endpoint $x = a \pm R$ **must be checked separately** (the Ratio Test is inconclusive there).

KEY FACT Finding the Interval of Convergence (Ratio Test Method)

Apply the Ratio Test to $a_n = c_n(x-a)^n$: compute $L(x) = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$. Solve $L(x) < 1$ for x — this gives the open interval and R . Then plug each endpoint back into the **original series** (not the ratio) and test it on its own merits (Divergence Test, p -series, Alternating Series Test, etc. — see Weeks 9-10).

KEY FACT Theorem 6.4: Term-by-Term Differentiation and Integration for Power Series

If $f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$ has radius of convergence $R > 0$, then for $|x-a| < R$,

$$f'(x) = \sum_{n=1}^{\infty} n c_n(x-a)^{n-1}, \quad \int f(x) dx = C + \sum_{n=0}^{\infty} \frac{c_n}{n+1}(x-a)^{n+1},$$

and **both** the differentiated and integrated series have the **same radius of convergence** R as the original (endpoint behavior can change, even though R does not).

KEY FACT Building New Power Series from $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, |x| < 1$

The geometric series is the master template. To represent a new function as a power series, manipulate it algebraically (substitute for x , factor out constants, split into partial fractions) until it looks like $\frac{1}{1 - (\text{something})}$, then substitute term-by-term. Combined with Theorem 6.4, this also lets you build series for functions whose derivative or antiderivative is a disguised geometric series (e.g. $\ln(1+x)$, whose derivative $\frac{1}{1+x}$ is geometric).

Worked Examples

EXAMPLE 1

interval of convergence, standard case

Find the radius and interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-3)^n}{n \cdot 4^n}$.

Solution. $\left| \frac{a_{n+1}}{a_n} \right| = \frac{|x-3|^{n+1}}{(n+1)4^{n+1}} \cdot \frac{n \cdot 4^n}{|x-3|^n} = \frac{n}{n+1} \cdot \frac{|x-3|}{4} \rightarrow \frac{|x-3|}{4}$. Need $\frac{|x-3|}{4} < 1 \iff |x-3| < 4$, so $R = 4$, open interval $(-1, 7)$. **Endpoint** $x = 7$: series becomes $\sum \frac{4^n}{n \cdot 4^n} = \sum \frac{1}{n}$ — diverges (p -series, $p = 1$). **Endpoint** $x = -1$: series becomes $\sum \frac{(-4)^n}{n \cdot 4^n} = \sum \frac{(-1)^n}{n}$ — converges (Alternating Series Test: terms $\rightarrow 0$, decreasing). Interval of convergence: $[-1, 7)$. ■

EXAMPLE 2

the “hard” factorial-ratio case

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{(3n)!}{n! (2n)!} x^n$.

Solution. $\left| \frac{a_{n+1}}{a_n} \right| = \frac{(3n+3)!}{(n+1)! (2n+2)!} \cdot \frac{n! (2n)!}{(3n)!} |x| = \frac{(3n+1)(3n+2)(3n+3)}{(n+1)(2n+1)(2n+2)} |x|$. Both numerator and denominator are cubic in n with leading coefficients 27 and 4, so the ratio $\rightarrow \frac{27}{4} |x|$. Need $\frac{27}{4} |x| < 1 \iff |x| < \frac{4}{27}$. **Radius of convergence** $R = \frac{4}{27}$. (Misconception flag: don’t stop at “it’s a ratio of factorials so $R = 0$ ” — factorial growth can cancel almost completely when it appears on both top and bottom; always simplify the ratio algebraically before taking the limit.) ■

EXAMPLE 3

representing a function as a power series, factor first

Represent $f(x) = \frac{1}{2-x}$ as a power series centered at 0, and find its interval of convergence.

Solution. Factor a 2 out of the denominator: $\frac{1}{2-x} = \frac{1}{2} \cdot \frac{1}{1-\frac{x}{2}} = \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{x}{2}\right)^n = \sum_{n=0}^{\infty} \frac{x^n}{2^{n+1}}$, valid when $\left|\frac{x}{2}\right| < 1$, i.e. interval of convergence $(-2, 2)$. ■

EXAMPLE 4

representing $\frac{1}{1-x^2}$, substitution

Represent $f(x) = \frac{1}{1-x^2}$ as a power series and find its interval of convergence.

Solution. Substitute x^2 for x in $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$: $\frac{1}{1-x^2} = \sum_{n=0}^{\infty} (x^2)^n = \sum_{n=0}^{\infty} x^{2n}$, valid for $|x^2| < 1 \iff |x| < 1$. Interval of convergence: $(-1, 1)$. ■

EXAMPLE 5

$\ln(1+x)$, integrate a known series — the “compute the derivative first” hint

Represent $f(x) = \ln(1+x)$ as a power series centered at 0.

Solution. $f'(x) = \frac{1}{1+x} = \frac{1}{1-(-x)} = \sum_{n=0}^{\infty} (-x)^n = \sum_{n=0}^{\infty} (-1)^n x^n$ for $|x| < 1$. Integrating term-by-term (Theorem 6.4):

$$f(x) = \int f'(x) dx = C + \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}.$$

Set $x = 0$: $f(0) = \ln 1 = 0 = C$, so $C = 0$. Reindex with $m = n+1$: $f(x) = \sum_{m=1}^{\infty} (-1)^{m-1} \frac{x^m}{m}$,

valid (at least) for $|x| < 1$ — the radius doesn’t change under integration, though the endpoint $x = 1$ needs a separate check (it turns out to converge there too, by AST, giving $\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \dots$, a classical fact you may quote but not prove here). ■

Practice Problems

Read the hint and the caution before you start each problem — not after.

Interval and radius of convergence

- Find the radius and interval of convergence of $\sum_{n=1}^{\infty} \frac{(-1)^n (x+2)^n}{n^2}$.

HINT Use the Ratio Test on $|(-1)^n (x+2)^n / n^2|$ to find R , then plug both endpoints $x = -1$ and $x = -3$ back into the original series separately.

CAUTION — before you solve Don't assume alternating behavior at one endpoint forces the other to diverge — check each endpoint on its own; here both converge because $p = 2$ gives absolute convergence regardless of sign.

Your working:

2. Find the radius and interval of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{n!}$. (No endpoint work needed — explain why, in one sentence, from the value of R .)

HINT Apply the Ratio Test to $x^n/n!$ and see what the limit of the ratio equals for every fixed x .

CAUTION — before you solve Endpoints only exist — and only need checking — when R is a finite positive number; don't try to 'check $x = \pm\infty$ ' when $R = \infty$.

Your working:

3. Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{n! x^n}{n^n}$.

HINT Set up the ratio $\left| \frac{a_{n+1}}{a_n} \right|$ and simplify $\left(\frac{n}{n+1} \right)^n$ using the standard limit that defines e .

CAUTION — before you solve $\left(\frac{n}{n+1}\right)^n$ tends to $1/e$, not 1 — don't cancel the fraction to 1 before taking it to the n -th power.

Your working:

Representing functions as power series

4. Represent $f(x) = \frac{1}{1+3x}$ as a power series centered at 0 and give its interval of convergence.

HINT Rewrite $\frac{1}{1+3x}$ as $\frac{1}{1-(-3x)}$ to match the geometric series form $\sum r^n$ with $r = -3x$.

CAUTION — before you solve Raise the ENTIRE term $-3x$ to the n -th power — writing $(-3)x^n$ instead of $(-3)^n x^n$ loses the alternating sign pattern.

Your working:

5. Represent $f(x) = \frac{x}{4+x^2}$ as a power series centered at 0 and give its interval of convergence. (Hint: factor a 4 out first, as in Example 3, then multiply by x .)

HINT Factor a 4 out of the denominator first to get a geometric form in $-x^2/4$, then multiply the whole series by the leftover $x/4$ at the end.

CAUTION — before you solve Don't forget the leftover factor of x (and the $\frac{1}{4}$) sitting outside the geometric series — it's easy to drop after substituting.

Your working:

6. Use the geometric series and Theorem 6.4 to represent $f(x) = \frac{1}{(1-x)^2}$ as a power series. (Hint: $\frac{1}{(1-x)^2}$ is the derivative of $\frac{1}{1-x}$.)

HINT Differentiate the closed form $\frac{1}{1-x}$ to get $\frac{1}{(1-x)^2}$, then differentiate its power series term by term using Theorem 6.4.

CAUTION — before you solve After differentiating $\sum x^n$ term by term, the sum correctly starts at $n = 1$ — don't relabel it to start at $n = 0$ without also shifting the exponent, or you'll include a nonsense term.

Your working:
